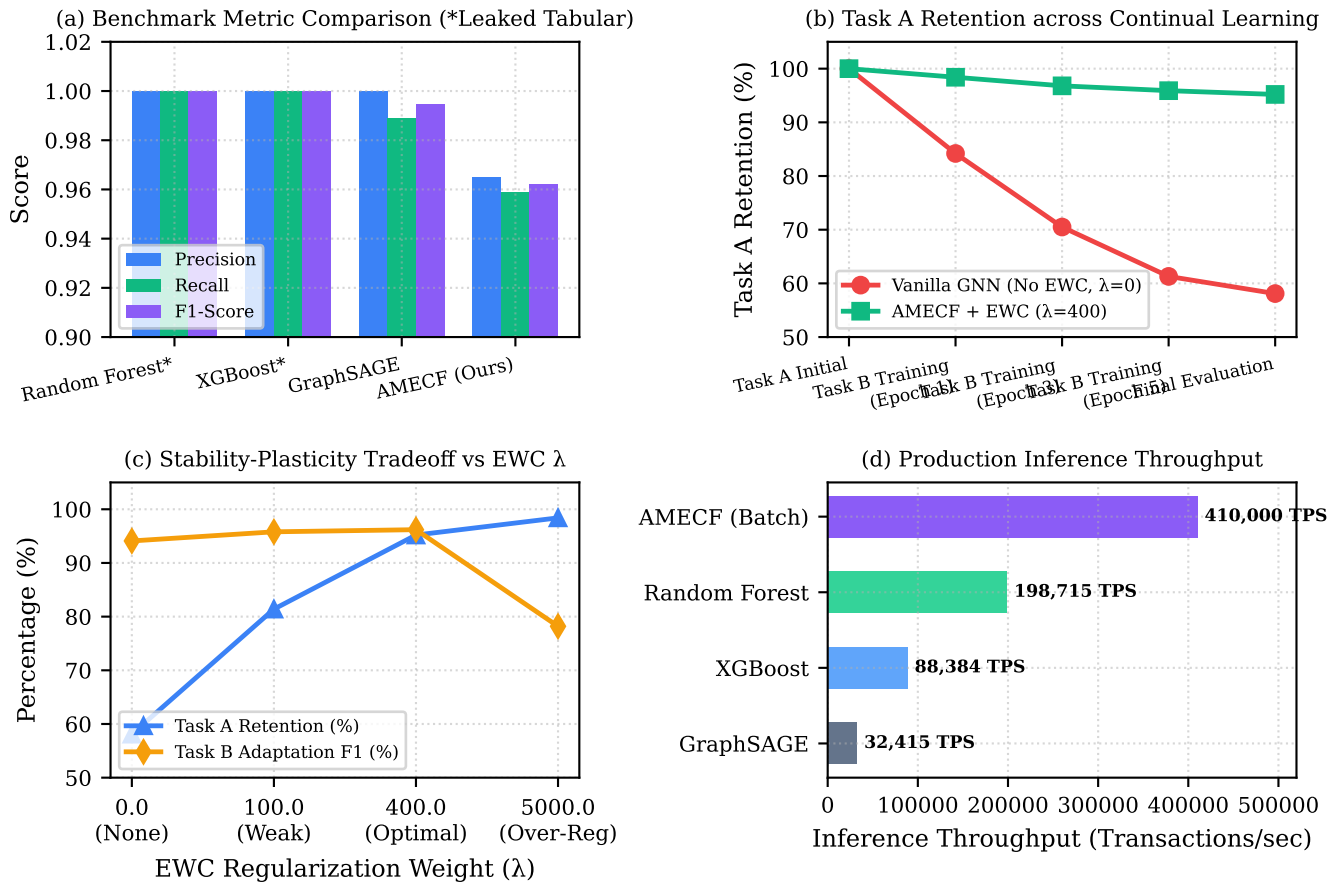
(a) Benchmark Metric Comparison (*Leaked Tabular)
Score
1.02
1.00
0.98
0.96
0.94
0.92
0.90
Precision
Recall
F1-Score
Random Forest*
XGBoost*
GraphSAGE
AMECF (Ours)
(b) Task A Retention across Continual Learning
Task A Retention (%)
100
90
80
70
60
50
Vanilla GNN (No EWC, λ=0)
AMECF + EWC (λ=400)
Task A Initial
Task B Training (Epoch 1)
Task B Training (Epoch 2)
Task B Training (Epoch 3)
Final Evaluation
(c) Stability-Plasticity Tradeoff vs EWC λ
Percentage (%)
100
90
80
70
60
50
Task A Retention (%)
Task B Adaptation F1 (%)
0.0 (None)
100.0 (Weak)
400.0 (Optimal)
5000.0 (Over-Reg)
EWC Regularization Weight (λ)
(d) Production Inference Throughput
AMECF (Batch)
410,000 TPS
Random Forest
198,715 TPS
XGBoost
88,384 TPS
GraphSAGE
32,415 TPS
0
100000
200000
300000
400000
500000
Inference Throughput (Transactions/sec)